

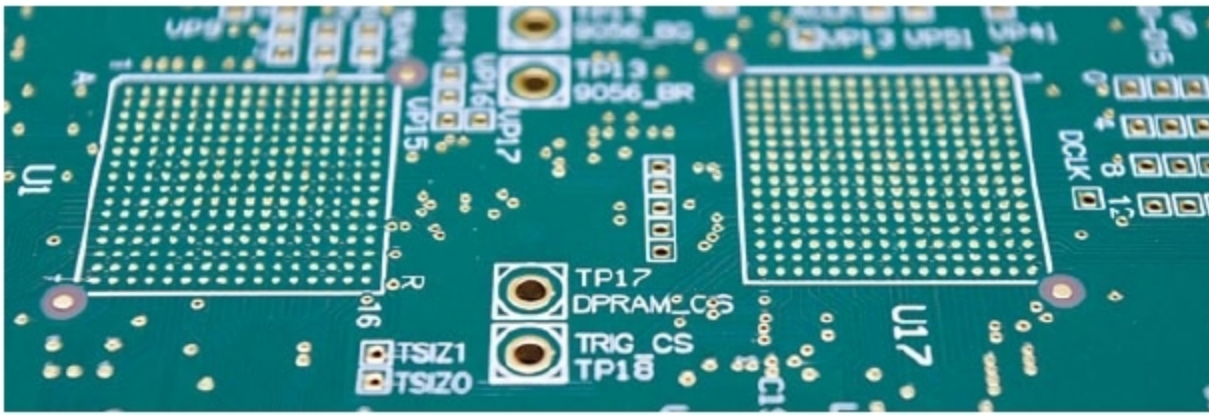


Fig. 1: High-density interconnect PCB (Credit: Millennium Circuits Ltd)

metals from e-waste to reuse them is great. This could reduce the pressures on mining companies to produce vast amounts of metal for the electronics industry. Biodegradable substrates are a good alternative, which is currently under close examination. Both of these methods could help reduce the impact of the electronics industry on the environment.

In the mobile telecom industry, interconnect and board dimensions are reducing in size rapidly. Nowadays, PCBs are utilising fewer but more complex components. Boards for networking and computer applications require more interconnect and plane layers. Understanding which tools will best support the needs of design is therefore necessary.

Electronics companies require best PCB design tools with effective libraries, infrastructure, design data and designers. These allow design teams working at different locations to work on the same project at a time.

Latest technology for PCB fabrication, design and interconnects

One of the upcoming trends is using PCBs as active system components.

Currently, PCBs are used as connecting components in electronics, relaying messages between active components so the complete device can work. Engineers are currently working on active PCBs, which will reduce the number of components in the PCBs while maintaining functionality.

Augmented reality (AR) and virtual reality (VR) could change the face of electronics design, as these are making rapid inroads in the consumer electronics world. Their potential application in PCB design is to address issues such as fitting electronic packages into unconventional shapes. This will ensure circuit connections work properly and reduce the time-consuming process of place-and-route.

The idea of training using AR with simulation software is also under exploration. In the near future, augmented PCBs could be used to reduce cost and wastage of materials used in current training methods. It could replicate magnetic and electric fields generated by a product as an alternative way to ensure that the product meets regulations. "Much like the integration of 3D into PCB design, AR is a solution to many problems of advanced technology," says Ben

Jordan, senior manager product and persona marketing, Altium.

As consumer demands change, so must technology. High-density interconnect (HDI) boards are one of the fastest growing technologies. These have higher density circuitry than traditional circuit boards. This provides designers the option to place more components on both sides of the PCB.

Multiple via processes allow designers to place components that are smaller even closer together. Decreased component size allows more input/output in smaller geometries. This results in faster transmission of signals and significant reduction in signal loss.

A PCB board camera is mounted directly onto a circuit board. It consists of a lens, aperture and image sensor. It is designed to take both digital pictures and videos, and can be mounted on any PCB, irrespective of size.

Board cameras have developed quickly to take high-resolution images and videos with ease. In the future, board cameras are expected to develop to create powerful solutions for both industry and consumer electronics.

Small board cameras have found important applications in the medical industry for non-invasive and minimally-invasive procedures. Pill-sized cameras allow doctors to take comprehensive videos and images from inside the digestive tract without invasive surgery. These small cameras can also be used to monitor homes and businesses for intruders.

3D polyethylene (PE) manufacturing is expanding the types of materials that can be used for circuits. It makes use of flexible and biodegradable sub-

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